

Inho Yoo

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EDUCATION

Kwansei Gakuin University | Hyogo, Japan

Bachelor of Science in Physics and Astronomy | *Expected March 2027*

- **Advisor / Senior Thesis Supervisor:** Prof. Shuji Matsuura
- **Relevant Coursework:** Astrophysics, Infrared Astronomy, X-ray Astronomy, Computational Physics

RESEARCH INTERESTS

Planetary Spectroscopy • Asteroid Surface Composition • Small Bodies • Remote Sensing • Space Resources • Multimodal Scientific Machine Learning

RESEARCH EXPERIENCE

Undergraduate Researcher, Matsuura Laboratory —
Kwansei Gakuin University

Project: *Data Analysis Pipeline and Spatial Distribution Mapping for Cosmic Ice Exploration*

- Analyzed the SPHEREx SPLICES target catalog, a 3.1 GB FITS dataset containing ~9 million background sources selected for infrared absorption spectroscopy of interstellar ice.
- Built a memory-efficient Python/Astropy pipeline using memory-mapped FITS access and random sampling of 10,000 sources to preserve large-scale distribution while reducing computational load.
- Mapped all-sky source positions in equatorial coordinates and interpreted the S-shaped concentration as the projected Galactic Plane; identified enhanced density toward the Galactic Center, consistent with SPLICES targeting of molecular-cloud-rich sightlines.
- Planned next phase: develop and validate Python spectral-fitting routines on public OSIRIS-REx Bennu infrared data to detect H₂O and hydrated-mineral absorption features before SPHEREx spectral data become available.

SELECTED RESEARCH PROJECT

Multimodal Machine Learning for Asteroid Compositional Mapping — *Research Proposal*

- Proposed a multimodal scientific machine-learning framework for compositional inference from heterogeneous infrared and X-ray remote-sensing observations of planetary surfaces.
- Designed a Moon → Bennu → Ryugu progression to study complete, noisy, and missing sensing modalities across different planetary data environments.
- Proposed validation of remote-sensing composition estimates against OSIRIS-REx and Hayabusa2 returned-sample measurements, linking spacecraft spectroscopy with laboratory ground truth.

TECHNICAL SKILLS

Programming: Python | **Scientific Computing:** Astropy, NumPy, Pandas, Matplotlib, PDS4 Tools |
Data Analysis: large-scale FITS processing, memory mapping, statistical sampling, spatial mapping, coordinate transformations | **Tools:** LaTeX, Visual Studio Code